LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

8 - MIMO Systems - Introduction

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Slide 1

TOPICS

MIMO CONTROL

Examples

(i) Driving a Car

Inputs: steering wheel angle, accelerator pedal angle

Outputs: velocity, direction$_x$, direction$_y$

(ii) Flying an aeroplane

Inputs: rudder, aileron, elevator...

Outputs: yaw, roll, slip....

(iii) Many others

- helicopter
- ship steering
- chemical process control
- macroeconomy
- autonomous vehicles ...

MIMO CONTROL ISSUES

(i) Extend SISO paradigm i.e. extend analysis tools and design methods

(ii) New problems with no SISO analogue, e.g. interaction of control signals such as aeroplane flight or turning a car around a tight carrier.

$\Rightarrow$design to achieve reduced or no interaction i.e. decoupling

MIMO Example

Quadruple Tank

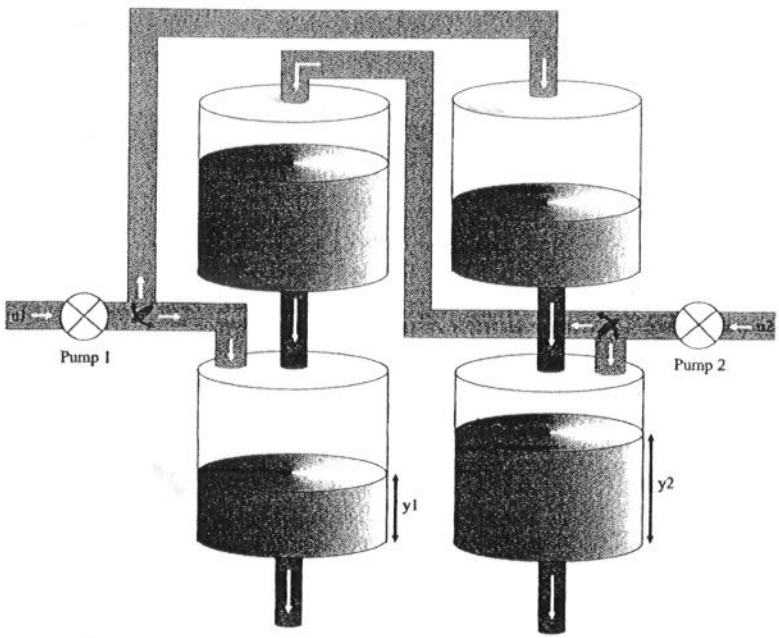

Figure 20.2. Schematic of a quadruple-tank apparatus

Source: Goodwin et al 2000

Physical modeling leads to the following (linearized) transfer function linking (u_1, u_2) *with* (y_1, y_2).

$$\mathbf{G}(s) = \begin{bmatrix} \dfrac{3.7\gamma_1}{62s+1} & \dfrac{3.7(1-\gamma_2)}{(23s+1)(62s+1)} \\ \dfrac{4.7(1-\gamma_1)}{(30s+1)(90s+1)} & \dfrac{4.7\gamma_2}{90s+1} \end{bmatrix} \qquad (20.3.48)$$

Where γ_1 *and* $(1-\gamma_1)$ *represent the proportion of the flow from pump 1 that goes into tanks 1 and 4, respectively (similarly for* γ_2 *and* $(1-\gamma_2)$*). Actually, the time constants change a little with the operating point, but this does not affect us here.*

MATHEMATICAL DESCRIPTION

There are 4 main ways to describe MIMO systems mathematically. We suppose m inputs $u_{mx1}(t); p$ outputs $y_{px1}(t)$

(i) Transfer function (TF)

or bivariate or pairwise ODE obtained by writing down pairwise the ODE's linking input j to output i. Leads to TF, $P_{ij}(s)$. Not always possible to do straightoff. The transfer function matrix is $P(s) = [P_{ij}(s)]$.

(ii) Matrix Fraction Description (MFD)

or multivariable ODE. Write down the ODEs relating all variables/signals together. Leads to

$$
\begin{aligned}
& (s^n I + D_1 s^{n-1} + \cdots + D_n)\overline{y}(s) \\
= \quad & (N_1 s^{n-1} + N_2 s^{n-2} + \cdots + N_n)\overline{u}(s) \\
\Rightarrow \overline{y} \quad = \quad & D_L^{-1}(s) N_L(s) \overline{u}
\end{aligned}
$$

This yields a left MFD of the TF matrix: $P(s) = D_L^{-1}(s) N_L(s)$

MATHEMATICAL DESCRIPTION - Continued

(i) State Space (SS)

Much as in the SISO case

$\dot{x} = Ax + B_{p\times m}u, y = C^T_{p\times n}x$

(iv) Polynomial Matrix Description (PMD)

Consists of a pair of equations:

ODEs linking partial state to inputs

$D(s)\overline{\xi} = N(s)\overline{u}$

Observation ODEs

$$\begin{aligned} \overline{y} &= C(s)\overline{\xi} + E(s)\overline{u} \\ \Rightarrow \quad \overline{y} &= C(s)D^{-1}(s)N(s)\overline{u} + E(s)\overline{u} \end{aligned}$$

When the partial state equation is "all pole"

$$D_R(s)\overline{\xi} = \overline{u}, \overline{y} = N_R(s)\overline{\xi}$$

We get a right MFD

$$\begin{aligned} \overline{y} &= N_R(s)D_R^{-1}(s)\overline{u} \\ \Rightarrow P(s) &= N_R(s)D_R^{-1}(s) \end{aligned}$$

Example 8.1

$$P(s) = \begin{bmatrix} \frac{3s+4}{(s+1)(s+2)} & \frac{1}{(s+2)(s+3)} \\ \frac{2(2s+3)}{(s+1)(s+3)} & \frac{2s+5}{(s+2)(s+3)} \end{bmatrix}$$

Can get impulse response (IR) by inverting partial fraction expansion of each element giving

$$IR(t) = \mathcal{H}(t) \begin{bmatrix} e^{-t} + 2e^{-2t} & e^{-2t} - e^{-3t} \\ e^{-t} + 3e^{-3t} & e^{-2t} + e^{-3t} \end{bmatrix}$$

Where $\mathcal{H}(t) =$ Heaviside step function

$$SS \to TF$$

MIMO state space description is a simple extension of SISO-SS

$$\begin{aligned}
\dot{x} &= Ax + Bu \\
&= Ax + (b_1, \cdots, b_m) \begin{pmatrix} u_1 \\ \vdots \\ \vdots \\ u_m \end{pmatrix} \\
&= Ax + \sum_{1}^{m} b_i u_i
\end{aligned}$$

SS $\rightarrow$ TF - Continued

$$
\begin{aligned}
y &= C^T x \\
&= \begin{pmatrix} c_1^T \\ \vdots \\ \vdots \\ c_p^T \end{pmatrix} x \\
\Rightarrow \begin{pmatrix} y_1 \\ \vdots \\ \vdots \\ y_p \end{pmatrix} &= \begin{pmatrix} c_1^T x \\ \vdots \\ \vdots \\ c_p^T x \end{pmatrix}
\end{aligned}
$$

SS →TF & LT

Again LT is a straight forward extension

$$
\begin{aligned}
s\overline{x} &= A\overline{x} + B\overline{u} + x_0 \\
\Rightarrow \overline{x} &= (sI - A)^{-1}B\overline{u} + (sI - A)^{-1}x_0 \\
\Rightarrow \overline{y} &= C^T\overline{x} \\
&= C^T(sI - A)^{-1}B\overline{u} + C^T(sI - A)^{-}x_0 \\
&= P(s)\overline{u} + IC \\
\Rightarrow P(s) &= C^T(sI - A)^{-1}B \\
&= \begin{pmatrix} c_1^T \\ \vdots \\ \vdots \\ c_p^T \end{pmatrix} (sI - A)^{-1}(b_1, \cdots, b_m) \\
&= [c_i^T(sI - A)^{-1}b_j] \\
&= [P_{ij}(s)]
\end{aligned}
$$

So as expected the TF linking input j to output i is $P_{ij}(s) = c_j^T(sI - A)^{-1}b_j$

TF → SS

•The story here is not so straightforward. We proceed in several stages.

•Introduce

$d(s) = s^r + d_1 s^{r-1} + \cdots + d_r$

=least common multiple of denominators in $P(s)$

•Example 8.2: By inspection of Example 8.1

$$\begin{aligned} d(s) &= (s+1)(s+2)(s+3) \\ &= s^3 + 6s^2 + 11s + 6 \end{aligned}$$

• By elementary algebra we can write

$$\begin{aligned} P(s) &= \frac{N(s)}{d(s)} \\ N(s) &= N_1 s^{r-1} + N_2 s^{r-2} + \cdots + N_r \end{aligned}$$

(N_i is a pxm matrix of polynomials)

•From this we can construct a SS representation / form / realization.

Example 8.3

•By inspection from Examples 8.1 , 8.2

$$\begin{aligned} N(s) &= \begin{bmatrix} (3s+4)(s+3) & s+1 \\ (4s+6)(s+2) & (2s+5)(s+1) \end{bmatrix} \\ &= \begin{bmatrix} 3s^2+13s+12 & s+1 \\ 4s^2+14s+12 & 2s^2+7s+5 \end{bmatrix} \\ &= s^2 \begin{bmatrix} 3 & 0 \\ 4 & 2 \end{bmatrix} + s \begin{bmatrix} 13 & 1 \\ 14 & 7 \end{bmatrix} + \begin{bmatrix} 12 & 1 \\ 12 & 5 \end{bmatrix} \end{aligned}$$

BLOCK CONTROLLER FORM

- Claim SS realization is given by

$$A_{mr\times mr} = \begin{bmatrix} -d_1 I_m & -d_2 I_m & \cdots & -d_r I_m \\ I_m & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & I_m & 0 \end{bmatrix}$$

$$B_{mr\times m} = \begin{bmatrix} I_m \\ 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix}, C^T_{p\times mr} = [N_1, \cdots, N_r]$$

BLOCK CONTROLLER FORM - Continued

• SS model

$$\begin{aligned}\dot{x}_c &= Ax_c + Bu \\ y &= C^T x_c \\ x_c &= \begin{bmatrix} x_{c,1} \\ \vdots \\ \vdots \\ x_{c,m} \end{bmatrix}, x_{c,i} \text{ are m-vectors}\end{aligned}$$

• $dim(x_c) = mr$

• Similarly we can construct a block observer form but with $\dim(x_0) = pr \neq mr$ (unless $p = m$)

This raises the question of 'correct' or minimal state dimension.

PROOF OF CONTROLLER FORM

Clearly

$\overline{y} = \sum_1^r N_i \overline{x}_i (x_i \equiv x_{c,i})$

Also

$$\overline{x}_1 = s\overline{x}_2 = s^2\overline{x}_3 = \cdots = s^{r-1}\overline{x}_r$$

$\Rightarrow$ in general $\overline{x}_i = s^{r-i}\overline{x}_r$

Thus

$$\begin{aligned}
\overline{y} &= \sum_1^r N_i \overline{\xi}_i \\
&= \sum_1^r N_i s^{r-i} \overline{\xi}_r \\
&= N(s)\overline{\xi}_r \\
N(s) &= \Sigma_1^r N_i s^{r-i}
\end{aligned}$$

PROOF OF CONTROLLER FORM - Continued

Next, the top SS equation gives

$$s\overline{x}_1 + \sum_1^r d_i \overline{\xi}_i = \overline{u}$$

$$\begin{aligned}
\Rightarrow \quad & \sum_0^r s^{r-i} d_i \overline{\xi}_r = \overline{u}; d_0 = 1 \\
\Rightarrow \quad & d(s)\overline{\xi}_r = \overline{u}, d(s) = \Sigma_1^r d_i s^{r-i} \\
\Rightarrow \quad & \overline{y}(s) = N(s)\overline{\xi}_r(s) \\
= \quad & N(s)\frac{\overline{u}(s)}{d(s)} \\
= \quad & P(s)\overline{u}(s)
\end{aligned}$$

as required.

NB: $\overline{\xi}_r(s)$ is called the partial state.

CONTROLLABILITY OBSERVABILITY

• Definitions are as in the SISO case and lead to the same criteria.

•A, B is controllable iff
$\mathcal{C}(A,B)_{n\times nm} = [B, AB, \cdots, A^{n-1}B]$ has rank n

The controllability index is the smallest q so that
$\mathcal{C}_q = (B, AB, \cdots, A^{q-1}B)$ has rank n

CONTROLLABILITY OBSERVABILITY
Continued

- C, A is observable iff

$$\mathcal{O}(C,A)_{pn\times n} = \begin{bmatrix} C^T \\ C^T A \\ : \\ : \\ C^T A^{n-1} \end{bmatrix} \text{ has rank } n$$

The observability index is the smallest q so that

$$\mathcal{O}_q = \begin{bmatrix} C^T \\ C^T A \\ : \\ : \\ C^T A^{q-1} \end{bmatrix}$$

has rank n

CONTROLLABILITY, OBSERVABILITY
Continued II

$\mathcal{C}(A,B)_{n\times nm} = (B, AB, \cdots, A^{n-1}B)$

Write out B column by column

$=$

$[b_1, \cdots, b_m, Ab_1, \cdots, Ab_m, \cdots, A^{n-1}b_1, \cdots, A^{n-1}b_m]$

The rank of a matrix is unaltered by interchanging columns.

So $\mathcal{C}(A,B)$ has same rank as

$$
\begin{aligned}
&\mathcal{C}_*(A,B) \\
&= [b_1, Ab_1, \cdots, A^{n-1}b_1, b_2, Ab_2, \cdots, A^{n-1}b_2, \cdots, b_m, Ab_
\end{aligned}
$$

$$
\begin{aligned}
&= [\mathcal{C}(A,b_1), \cdots, \mathcal{C}(A,b_m)] \\
&= [\mathcal{C}_1, \cdots, \mathcal{C}_m]
\end{aligned}
$$

CONTROLLABILITY, OBSERVABILITY Continued III

Now the rank of $\mathcal{C}(A,B)$ is the same as the rank of the finite controllability Grammian

$$G(A,B)_{n\times n} = \mathcal{C}_*(A,B)\mathcal{C}_*^T(A,B)$$

And

$$\begin{aligned} G(A,B)_{n\times n} &= [\mathcal{C}_1,\cdots,\mathcal{C}_m]\begin{pmatrix} \mathcal{C}_1^T \\ \vdots \\ \vdots \\ \mathcal{C}_m^T \end{pmatrix} \\ &= \sum_1^m \mathcal{C}_i\mathcal{C}_i^T \end{aligned}$$

So $G(A,B)$ is singular iff $\mathcal{C}_i\mathcal{C}_i^T$ are singular $i = 1.....m$

CONTROLLABILITY OBSERVABILITY IV

Proof: If $G(A, B)$ is singular there exists θ such that

$$\begin{aligned} G(A,B)\theta &= 0 \\ \Rightarrow \theta^T G(A,B)\theta &= 0 \\ \Rightarrow \sum_1^m \theta^T \mathcal{C}_i \mathcal{C}_i^T \theta &= 0 \\ &= \sum_1^m \left\|\mathcal{C}_i^T \theta\right\|^2 \\ \Rightarrow \left\|\mathcal{C}_i \theta\right\|^2 &= 0, i = 1, \cdots, m \end{aligned}$$

So each $\mathcal{C}_i\mathcal{C}_i^T$ must be singular i.e. each $\mathcal{C}_i$must be of rank $< n$ i.e. lack of controllability must hold for each input.

Notes

(i) Similar result holds for observability.

(ii) For SS-modal form, loss of controllability means a row of B$_m$ is zero.

SS-Modal Form

[not to be confused with TF-Modal Form = GILBERT's form]

Given a SS model we can convert it to SS- modal form as in the SISO case.

Given A,B,C then assume A has distinct eigenvalues

$$|sI - A| = \pi_1^r (s - \lambda_i)$$

And let q_i, p_i be the corresponding left and right eigenvectors with

$$\begin{aligned} Q^T P &= I = PQ^T \\ A &= P \Lambda Q^T \\ Q^T &= \begin{bmatrix} q_1^T \\ \vdots \\ \vdots \\ q_m^T \end{bmatrix} \end{aligned}$$

SS-Modal Form Continued

Then transform the state to

$$
\begin{aligned}
x_M &= Q^T x \\
\dot{x}_M &= Q^T \dot{x} \\
&= Q^T A P Q^T x + Q^T B u \\
&= \Lambda x_M + B_M u \\
B_M &= Q^T B = \text{modal control matrix} \\
y &= C^T x \\
&= C^T P Q^T x \\
&= C_M^T x_M \\
C_M &= P^T C = \text{modal observation matrix}
\end{aligned}
$$

So, $A, B, C \rightarrow \Lambda, B_M, C_M$

Controllability and Observability and SS-modal form

•In the SISO case: Modal-SS form is uncontrollable iff one entry in the β vector is zero

• So in the MIMO case since each controllability matrix must be singular. Modal SS form is uncontrollable iff one row of B_M is zero.

• Similarly Modal SS form is unobservable iff one column of the modal observation matrix is zero.

DECOMPOSITION THEOREM

Denote rows of B_M by r_k^T

$$B_M = \begin{pmatrix} r_1^T \\ \vdots \\ \vdots \\ r_n^T \end{pmatrix}$$

Denote columns of C_M^T by χ_k

$$C_M^T = [\chi_1, \cdots, \chi_n]$$

We can thus classify the modes in four ways.

Mode k is

$c, 0$	if	$r_k \neq 0, \chi_k \neq 0$	no null rows of B_M no null column of C_M^T
$\overline{c}, 0$	if	$r_k = 0, \chi_k \neq 0$	null row of B_M
$c, \overline{0}$	if	$r_k \neq 0, \chi_k = 0$	null column of C_M^T
$\overline{c}, \overline{0}$	if	$r_k = 0, \chi_k = 0$	null row of B_M null column of C_M^T

TF from SS-modal form

$$
\begin{aligned}
\bullet P(s) &= C^T(sI-A)^{-1}B \\
&= C_M^T(sI-\Lambda)^{-1}B_M \\
&= (\chi_1,\cdots,\chi_n)(sI-\Lambda)^{-1}\begin{pmatrix} r^T \\ \vdots \\ \vdots \\ r_n^T \end{pmatrix} \\
&= \sum_1^n \frac{\chi_k r_k^T}{s-\lambda_k}
\end{aligned}
$$

• Now split the sum in four pieces according to decomposition theorem:

$$
\begin{aligned}
P(s) &= \sum_{k\in co}\frac{\chi_k r_k^T}{s-\lambda_k} + \sum_{k\in c\bar{o}}\frac{\chi_k r_k^T}{s-\lambda_k} \\
&+ \sum_{k\in \bar{c}o}\frac{\chi_k r_k^T}{s-\lambda_k} + \sum_{k\in \overline{co}}\frac{\chi_k r_k^T}{s-\lambda_k} \\
&= \sum_{k\in co}\frac{\chi_k r_k^T}{s-\lambda_k} + 0 + 0 + 0
\end{aligned}
$$

TF from SS-modal form Continued

•Note

Each $\chi_k r_k^T$ has rank 1 so this expression is not be confused with Gilbert's modal decomposition.

•# terms in sum = SS dimension. A minimal realization is a SS form of minimum dimension satisfying $P(s) = C^T(sI - A)^{-1}B$.

So A, B, C minimal iff both controllable and observable

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EXAMPLE 8.3

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -7 & -8 & -12 & 0 \\ 0 & 0 & 0 & 1 \\ -5 & -5 & -12 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 6 & 0 \\ -42 & 12 \\ 5 & 0 \\ -35 & 11 \end{bmatrix}$$

$$C = \frac{1}{6}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} ; \; D = 0$$

Find Modal decomposition and check controllability observability

```
(1) Type a,b,c,d into Matlab
(2) Form SS system
    ex8a=ss(a,b,c,d);
(3) use modal.m
    modal(ex8a);

a =

                   x1          x2          x3          x4
        x1         -4           0           0           0
        x2          0          -3           0           0
        x3          0           0          -2           0
        x4          0           0           0          -1


b =

                   u1          u2
        x1     154.62     -247.39
        x2     114.02     -347.75
        x3     14.534     -145.34
        x4 -2.1316e-013    23.335


c =

                   x1          x2          x3          x4
        y1   0.032338   -0.043853    0.068802     0.11785
        y2   -0.12935     0.13156     -0.1376    -0.11785


d =

                   u1          u2
        y1          0           0
        y2          0           0

Continuous-time model.
```

Note : u_1 is not controllable ; u_2 is controllable
y_1, y_2 are observable system

```
function [qt,p,lam]=modal(ssys)  % copyright V Solo 2001
%==============================================================
%  COMMAND  :  [qt,p,lam]=modal(ssys)
%   ACTION  :  modal decomposition of SS model
%
%   INPUTS  :  ssys = SS system (created by ssys=SS(a,b,c,d))
%
%  OUTPUTS  :  modal decomposition of a,b,c,d system
%==============================================================
[a,b,c,d]=ssdata(ssys);
[p,lam]=eig(a);
qt=inv(p);
bm=qt*b;
cm=c*p;
sysm=ss(lam,bm,cm,d)
```

PBH Tests

•These tests are straightforward extensions from SISO setting

•A, B is controllable iff there is no left evec of A orthogonal to all columns of B i.e. there is no q with

$$q^T A = \lambda q^T, q^T B = 0$$

NB

$$\begin{aligned} q^T B &= q^T (b_1, \cdots, b_m) \\ &= (q^T b_1, \cdots, q^T b_m) \\ &= 0 \\ \Rightarrow q^T b_i &= 0, i = 1, \cdots, m \end{aligned}$$

PBH Tests Continued

•C, A is observable iff there is not a right evec of$A$ orthogonal to all rows of C^T i.e. there is no p_R with

$Ap_R = \lambda p_R$ and $C^T p_R = 0$

NB

$$
\begin{aligned}
C^T p_R &= \begin{pmatrix} c_1^T \\ \vdots \\ \vdots \\ c_p^T \end{pmatrix} p_R \\
&= \begin{pmatrix} c_1^T p_R \\ \vdots \\ \vdots \\ c_p^T p_R \end{pmatrix} \\
\Rightarrow c_i^T p_R &= 0, i = 1, \cdots, p
\end{aligned}
$$

TF to SS via Gilbert's Method

Assume $d(s)$ has distinct roots.

$d(s) = \Pi_1^r (s - \mu_i)$

then we can do a partial fraction expansion.

$$
\begin{aligned}
P(s) &= \frac{N(s)}{d(s)} = \sum_1^r \frac{R_i}{s - \mu_i} \\
R_i &= \lim_{s \to \mu_i} P(s)(s - \mu_i)
\end{aligned}
$$

Let ρ_i =rank(R_i)

Then we can decompose

$$R_{i\,(p \times m)} = C_{i\,(p \times \rho_i)} B_{i\,(\rho_i \times m)}$$

TF to SS via Gilbert's Method Continued

Proof: BY SVD

$$R_i = \sum_{1}^{min(p,m)} u_t v_t^T \sigma_t$$

where we suppose the singular values are ordered $\sigma_1 \geq \sigma_2 \geq, \cdots$

Since rank $(R_i) = \rho_i$ there are only ρ_i non zero singular values so

$$\begin{aligned} R_i &= \sum_{1}^{\rho_i} u_t v_t^T \sigma_t \\ &= (u_1, \cdots, u_{\rho_i}) \begin{pmatrix} \sigma_1 v_1^T \\ \vdots \\ \vdots \\ \sigma_{\rho_i} v_{\rho_i}^T \end{pmatrix} \\ &= C_i B_i \end{aligned}$$

where C_i, B_i each have rank ρ_i

Example 8.4 - continuing examples 8.1 - 8.3

By inspection

$$\mu_1, \mu_2, \mu_3 = -1, -2, -3$$

Thus

$$\begin{aligned} R_1 &= \lim_{s \to -1} P(s)(s+1) \\ &= \lim_{s \to -1} \begin{bmatrix} \frac{3s+4}{s+2} & \frac{s+1}{(s+2)(s+3)} \\ \frac{2(2s+3)}{s+3} & \frac{(s+1)(2s+5)}{(s+2)(s+3)} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \end{aligned}$$

Similarly we find

$$R_2 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

$$R_3 = \begin{bmatrix} 0 & -1 \\ 3 & 1 \end{bmatrix}$$

And by inspection $\rho_1, \rho_2, \rho_3 = 1, 2, 2$

GILBERT's SS FORM

The SS form is (Gilbert's Form)

$$\left[\begin{array}{ccccc|c} \mu_1 I_{\rho_1} & 0 & 0 & \cdots & 0 & B_1 \\ 0 & \mu_2 I_{\rho_2} & \cdot & \cdots & \cdot & \cdot \\ 0 & \vdots & \cdot & & \cdot & \cdot \\ \vdots & \vdots & & & & \cdot \\ 0 & \cdot & \cdot & & \mu_r I_{\rho_r} & B_r \\ \hline C_1 & \cdot & \cdot & & C_r & \end{array}\right]$$

This SS form has order $n_{\text{min}} = \sum_1^r \rho_i$

- Example 8.5

$n_{\text{min}} = 1 + 2 + 2 = 5$

whereas $n_{BC} = rm = 3 \times 2 = 6[= n_{BO}]$

In fact we will see that the minimal order is $n_{\text{min}} = \sum_1^r \rho_i$ in this case.

Gilbert Form & Minimality

- The Gilbert SS form corresponding to a given TF is minimal

- Proof

A_G is symmetric so left evecs=right evecs. Suppose the SS form is not controllable then there is an evec of A_G , say q_G with eigenvalue μ such that $q_G^T B_G = 0$.

Since $A_G q_G = \mu q_G$ we see q_G must have the form

$$q_{G_{n\times 1}} = \begin{bmatrix} a_1(\rho_1 \times 1)) \\ 0(\rho_2 \times 1) \\ : \\ 0(\rho_n \times 1) \end{bmatrix}$$

Gilbert Form & Minimality Continued

$$\begin{aligned}
\Rightarrow 0 &= q_G^T B_G \\
&= (a_1^T, 0^T, \cdots, 0^T) \begin{bmatrix} B_1 \\ \vdots \\ \vdots \\ B_r \end{bmatrix} \\
&= a_1^T B_1
\end{aligned}$$

But now we have a ρ_1-vector, a_1 which nulls B_1. This contradicts B_1 being of rank ρ_1. So the SS form must be controllable.

Argument for observability is similar.